# Selective Recovery for Multilingual ASR: Candidate Acquisition and Acceptance

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***Abstract*—We separate fallback acquisition from acceptance using a compression trigger and a fixed four-feature selector. On a locked confirmation panel of 7,500 Common Voice 27.0 (CV27) recordings, selective recovery lowers macro word error rate (WER) by 2.26 percentage points (pp) over baseline rescoring and matches unconditional recovery's error counts in every seed. Retrospective CV27 TEST comparisons show 99.98% retention of macro-weighted positive edit gain. Reselection improves unscreened fallbacks; direct adoption can outperform the fixed selector on internally screened Hugging Face (HF) outputs. Matching acceptance to HF banks with edit-gap weighting yields the lowest exploratory CV27 WER of 31.44%. Across three checkpoints, selective acquisition saves up to 5.15% latency within reranked pipelines. Native HF remains faster but incurs 1.46 pp higher WER on FLEURS and over 2.1 pp on CV27. Code is available at https://github.com/Zhifan-Pan/ISCAS2027.**

***Index Terms*—multilingual speech recognition, selective recovery, candidate acquisition, N-best reranking, confidence estimation, conditional computation**

## I. Introduction

Multilingual ASR encounters routine errors and catastrophic repetition loops. Additional decoding helps only when it produces a better candidate that the scoring rule accepts. Separating acquisition from acceptance reveals when recovery helps or wastes computation.

Multilingual N-best reranking [1], [2], discriminative rescoring [3]–[5], voting [6] and confusion networks [7] exploit existing hypotheses. Generative correction expands banks [8]–[11], and risk-based pruning and ASR sampling allocate computation [12], [13]. Conditional computation uses recognizer cascades [14], selective passes [15], encoder proxies [16], selective correction [17] or language-specific distillation [18], but confidence need not predict the benefit of deferral [19].

We distinguish routine beam rescoring from sparse repetition recovery. We ask when bank expansion helps and when reselection helps or harms a screened fallback. The locked panel isolates acquisition; retrospective controls examine acceptance. We compare expected risk, pairwise ranking and minimum Bayes risk on matched banks [20]–[22]. Native HF trades higher WER for faster decoding.

## II. Controlled Acquisition and Acceptance

### A. Recording-level acquisition

We normalize and deduplicate five beams per window $i$ into $B_i$, retaining first occurrences and sources. With raw top beam $b_i$ and fallback $g_i$, $F_i = B_i \cup \{g_i\}$ contains at most six candidates. Before recovery, $\mathrm{CR}_{\mathrm{text}}$ divides top-beam UTF-8 byte length by zlib-compressed length, with zero for empty strings. The locked gate $z_i = \mathbf{1}\{\mathrm{CR}_{\mathrm{text},i} > 2.4\}$ adopts Whisper's standard text compression threshold [29] and gives $z^{\mathrm{rec}} = \max_i z_i$. Any trigger expands every window, even singletons (Fig. 1), as a conservative control preserving recording-level invocation. On single-window CV27 audio, $z^{\mathrm{rec}} = z_i$; Section IV-D examines local routing for multi-window FLEURS.

Recovery follows Hugging Face temperatures $0, 0.2, \ldots, 1$ with compression, log-probability and no-speech thresholds $1.35$, $-1$ and $0.6$. The internal check compresses serialized token-ID bytes [23]; the outer gate uses decoded UTF-8 text before requesting recovery. Its byte ratio reflects script encoding as well as repetition. Native HF uses the same configuration with conditional retries and returns the final transcript without N-best rescoring.

### B. Four-feature shared acceptance

For features $f(y) = (a, p, t, r)$, $a$ is the mean teacher-forced log probability over $T$ body tokens without prefix, padding or end tokens, and $t = \log(1 + T)$. XLM-R-base pseudo-log-likelihood $p$ [24], [25], or PLL, averages individually masked nonspecial subword log probabilities in centered contexts of at most 510 subwords of the original string, rather than within-word masking [26]. Empty outputs have no body tokens or PLL targets and map to $f(\emptyset) = (0, 0, 0, 0)$, but normalization alone need not yield zero features.

For normalized words $w_{1:M}$, the sequence-wide overlapping count is $c_y(u) = \sum_{k=1}^{M-n+1} \mathbf{1}\{w_{k:k+n-1} = u\}$. We measure repetition by

$$r_{n,h}(y) = \frac{\sum_{j=1}^{M-n+1} \mathbf{1}\{c_y(w_{j:j+n-1}) \geq h\}}{\max(1, M - n + 1)}. \quad (1)$$

We use $r(y) = r_{4,3}(y)$, which is zero for $M < 4$. Repetitive sequences also motivate unlikelihood training [27].

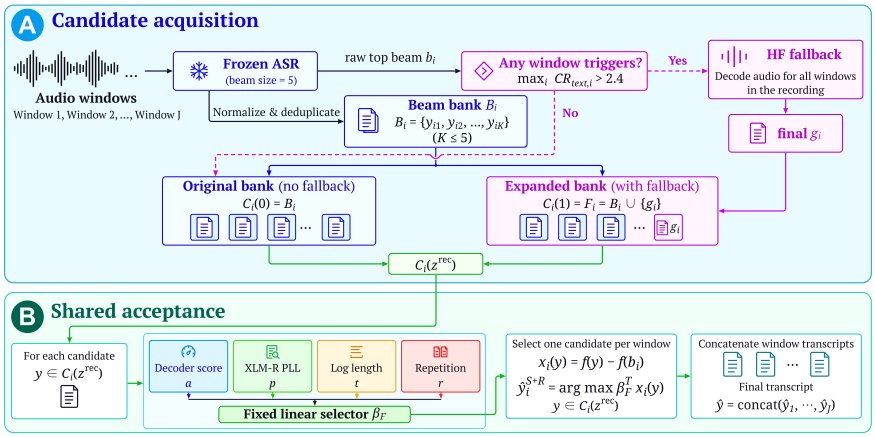



Fig. 1. Acquisition (A) expands every window's candidate bank if any window triggers. Acceptance (B) applies the same fixed four-feature selector to the original or expanded bank. Selected window transcripts are then concatenated.

Centering $x_i(y) = f(y) - f(b_i)$ preserves rankings and softmax probabilities. The empty output provides an implicit abstention baseline with score $-\beta_F^\top f(b_i)$. Feature-wise root-mean-square (RMS) scales $s \in \mathbb{R}^4_{>0}$ use training-fold pairs $\{(i,k) : i \in \bigcup_{\ell=1}^{L} \mathcal{I}_\ell,\ y_{ik} \in C_i \setminus \{b_i\}\}$. Scales below $10^{-12}$ become one. With reference $y_i^*$ and word-edit count $E$, define $\Delta_{ik} = E(b_i, y_i^*) - E(y_{ik}, y_i^*)$ and $\pi_{ik} = \text{softmax}_k(\theta^\top \text{diag}(s)^{-1} x_i(y_{ik}))$. Both softmax and loss cover $k \in \{1, \ldots, |C_i|\}$, including $b_i$ with $x_i(b_i) = 0$ and $\Delta_{i,b_i} = 0$. Expected-risk fitting [20] minimizes

$$\mathcal{L}(\theta) = -\frac{1}{L} \sum_{\ell=1}^{L} \frac{\sum_{i \in \mathcal{I}_\ell} \sum_{k=1}^{|C_i|} \pi_{ik} \Delta_{ik}}{N_\ell} + \lambda \|\theta\|_2^2. \quad (2)$$

Here $\mathcal{I}_\ell = \{i : \text{lang}(i) = \ell\}$ indexes fitting windows, $N_\ell$ counts reference words and $L = 5$.

The shared weights $\beta_F = \text{diag}(s)^{-1}\theta$ are fitted per run on greedy-expanded DEV and fixed across sources, with ties broken by bank order. With $\hat{y}_i^B = \text{argmax}_{y \in B_i} \beta_F^\top x_i(y)$, $C_i(0) = B_i$ and $C_i(1) = F_i$, the policies are

$$\hat{y}_i^{S+R} = \underset{y \in C_i(z^{\text{rec}})}{\text{argmax}}\ \beta_F^\top x_i(y),$$
$$\hat{y}_i^{\text{Direct}} = \begin{cases} \hat{y}_i^B, & z^{\text{rec}} = 0, \\ g_i, & z^{\text{rec}} = 1. \end{cases} \quad (3)$$

Policy B reranks $B_i$, and N+R requests and reranks $F_i$ unconditionally. S+R and Direct share acquisition and use B off-trigger, but Direct bypasses selection and PLL on triggers. Window transcripts are concatenated for full-reference scoring.

### C. Matched acceptance controls

G4 keeps $\beta_F$. On HF-expanded DEV banks, H4 refits four-feature expected risk and H3 omits PLL with $(a, t, r)$. HR4 fits RankNet [21] probabilities $\sigma(\beta^\top(x_{ij} - x_{ik}))$ for $E_j < E_k$, with $E_j = E(y_{ij}, y_i^*)$, $x_{ij} = x_i(y_{ij})$ and logistic $\sigma$. Language-reference normalization and equal total bank weight are retained. HWR4 weights pair losses by $w_{jk} = |E_j - E_k|/c$, where $c$ is the training-fold mean edit gap under the original pair weights.

TABLE I
WER (%) AND PP CONTRASTS BEFORE ROUNDING. CHECKS DENOTE HF ACQUISITION, CR GATING AND R RESELECTION (†: OFF-TRIGGER ONLY).

| Path | HF | CR | R | Scorer | FLEURS | CV27 |
|---|---|---|---|---|---|---|
| Raw beam-5 top-1 | – | – | – | – | 18.90 | 36.41 |
| Native HF [23] | ✓ | – | – | – | 18.55 | 33.80 |
| B | – | – | ✓ | G4 | 17.45 | 33.98 |
| N+R | ✓ | – | ✓ | G4 | 17.09 | 31.68 |
| S+R | ✓ | ✓ | ✓ | G4 | 17.09 | 31.68 |
| Direct | ✓ | ✓ | ✓† | G4 | 17.09 | 31.46 |
| H3Direct | ✓ | ✓ | ✓† | H3 | 18.68 | 33.64 |
| **A. Reference-assisted TEST oracle bounds** | | | | | | |
| Original-bank oracle | | | | | 16.19 | 30.19 |
| Full-HF oracle | | | | | 15.83 | 27.68 |
| CR-routed oracle | | | | | 15.83 | 27.68 |
| S+R gap to routed oracle, pp | | | | | 1.26 | 4.00 |

| **B. Locked CV27 confirmation (7,500 recordings)** | | | |
|---|---|---|---|
| | B | N+R = S+R | Gain |
| Macro-average | 35.02 | 32.75 | 2.26 |
| Contrast | | ΔWER | 95% upper bound |
| S+R − B | | -2.26 | -1.32 |
| S+R − N+R | | 0.00 | 0.00 |

Connected clients and reference texts define five folds per language. Held-fold macro WER selects among 49 log-spaced penalties from $10^{-4}$ to 1, preferring larger values on ties. L-BFGS-B refits full DEV from zero. Singletons enter WER denominators but give no pairs; padding is excluded.

MBR [22] selects $\hat{y}_i = \text{argmin}_{y \in C_i} \sum_{y' \in C_i} q_i(y') E(y, y')$. U-MBR uses $q_i(y') \propto 1$ and D-MBR uses $q_i(y') = \text{softmax}(Ta/\tau)_{y'}$. DEV WER selects $\tau \in \{0.1, 0.3, 1, 3, 10\}$, favoring larger values on ties; candidate ties retain bank order. *Complete* policies use each acceptor on both banks; *triggered* comparisons keep G4 off-trigger with full-panel denominators.

## III. Data and Evaluation

We adapt Whisper-small [29] on 44.60 h of FLEURS TRAIN [28] across Arabic, Persian, Polish, Turkish and Italian using LoRA [30] over three random seeds (4,656 updates, rank and scale four, zero dropout across all attention projections). LoRA training excludes CV27 TRAIN. CV27 DEV is used to fit selectors and check recovery with the fixed gate before confirmation. Evaluation on FLEURS TEST (13.07 h; 3,665

recordings) and CV27 TEST (83.00 h; 58,514 recordings) [31] is retrospective because both splits were accessed during development.

The locked confirmation panel has 7,500 CV27 TRAIN recordings (1,500 per language) with no previously accessed audio or reference texts. Predictions, stopping rules and the one-sided 0.10 pp noninferiority margin were locked before reference evaluation. We use 10,000 synchronized bootstrap draws of speaker components within languages. Speakers recurring across splits limit generalization to this sampling frame. Confirmation covers the original G4 acquisition policy; later gates, refits and window policies remain retrospective.

We weight costs by inverse sampling probabilities for 255 FLEURS and 472 CV27 recordings. The sample covers all first-checkpoint triggers or Native HF output changes and ten ordinary recordings per language and duration quintile. Study A measures four paths across three checkpoints; Study B measures seven on one. Later-checkpoint omissions comprise five FLEURS and 207 CV27 triggers (0.05% and 0.12% of seed–recordings). Triggered N+R and S+R execute identical operations, though sampling shares do not bound timing variation. Study B covers all its triggers and corroborates Study A's savings (5.18% and 4.54% versus 5.15% and 4.57%). Both use three randomized paired rounds with each recording block on one of four NVIDIA A40 (48 GB) GPUs, resident models, within-path encoder and first-search reuse, and 32-mask PLL batches. Waveform-to-output timing excludes loading, warmup and interfered blocks. Peak CUDA memory is the maximum per-process allocation in MiB over calls and rounds. Session-conditional intervals use 4,000 paired stratified draws with finite-population correction; Study A averages checkpoints with synchronized draws.

Known-language decoding uses bfloat16, five beams, length penalty one, native suppression and a 444-token cap in Transformers 4.57.6. At 16 kHz, $N$ samples give $J = \lceil N/480000 \rceil$ windows bounded by $\lfloor jN/J \rfloor$ without previous-window text. Deduplication, WER and $r$ share NFC normalization, digit unification, punctuation-to-space, Persian mapping and Turkish casing [32], with fixed source RNG mappings. WER averages five language aggregates and three seeds; calls pool seed–recordings. TEST intervals use 2,000 paired client/reference-component draws within languages, synchronized across seeds with fits and source draws frozen. FLEURS comparisons are descriptive because component metadata are unverified.

## IV. Results

### A. Acquisition with fixed acceptance

On the locked confirmation panel, S+R gains 2.26 pp over B, with seed gains of 0.20 to 4.54 pp (Table I). Matching N+R's error counts in every seed meets the prespecified 0.10 pp margin but does not guarantee zero loss under distribution shift. Of 61 triggers, 49 improve edits and two worsen them.

Retrospective evaluation yields 32.71% WER for Direct versus S+R's locked 32.75%. Both beat B and Native HF's retrospective 34.87% by over 2.1 pp. Direct's unrounded difference from S+R is $-0.05$ pp with a paired speaker-bootstrap 95% interval from $-0.19$ to $+0.04$ pp.

TABLE II
Complete-policy TEST WER (%) and gain B−S (pp) on HF banks. Checks denote PLL, pairwise P and edit-gap weighting W.

| Acceptor | PLL | P | W | FLEURS | | | CV27 | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | B | S | Gain | B | S | Gain |
| G4 [20] | ✓ | – | – | 17.45 | 17.09 | 0.37 | 33.98 | 31.68 | 2.30 |
| H4 [20] | ✓ | – | – | 17.46 | 17.09 | 0.37 | 33.97 | 31.52 | 2.45 |
| HR4 [21] | ✓ | ✓ | – | 17.51 | 17.14 | 0.37 | 34.02 | 32.30 | 1.72 |
| H3 [20] | – | – | – | 19.05 | 18.68 | 0.37 | 36.15 | 33.77 | 2.38 |
| HWR4 [21] | ✓ | ✓ | ✓ | 17.47 | 17.10 | 0.37 | 33.97 | 31.44 | 2.53 |
| U-MBR [22] | – | – | – | 19.14 | 19.14 | 0.00 | 36.91 | 36.88 | 0.03 |
| D-MBR [22] | – | – | – | 18.92 | 18.75 | 0.17 | 36.09 | 34.48 | 1.60 |

| | | |
|---|---|---|
| *A. Original family: CV27 ΔWER (individual 97.5% CI)* | | |
| Triggered HR4 − H4 | +0.72 | [+0.40, +1.16] |
| Complete H3 − H4 | +2.25 | [+2.16, +2.34] |
| *B. Later two-comparison family* | | |
| Triggered U-MBR − H4 | +2.51 | [+1.93, +3.18] |
| Triggered HWR4 − H4 | -0.08 | [-0.15, -0.03] |

On TEST, S+R retains all ten beneficial FLEURS instances at 0.09% calls and 370 of 373 CV27 instances at 0.30%, missing five edits. Language-reference normalization gives 99.98% macro-weighted positive edit gain retention. N+R selects HF on one FLEURS and 13 CV27 nontriggers where S+R keeps B, with nearly identical WER. Reference-assisted oracles minimize edits over concatenated original, expanded or routed banks. Routing changes this bound by under 0.01 pp against S+R's 4.00 pp CV27 gap. In retrospective TEST replay, the word gate $r_{4,3}(b_i) > 0$ lowers FLEURS gain retention from 100% to 93.34% (0.06% calls; 99.21% at 0.18% on CV27), whereas token-ID compression $> 1.35$ more than doubles FLEURS calls to 0.20% (0.31% on CV27) for 100%/99.99% retention, supporting the text gate as a parsimonious operating point on these banks.

### B. Interaction with the fallback source

S+R beats Direct on unscreened fallbacks by 0.53 pp with greedy recovery and 0.43 pp with one draw at temperature 0.2. HF screens retries by compression, log-probability and no-speech criteria. Direct then leads G4 by 0.22 pp, with a 95% interval from 0.11 to 0.37 pp, concentrated in checkpoint one ($-0.68$ pp versus $+0.00$ and $+0.02$ pp). There, greedy-fitted G4 retains repetitive beams whose decoder/PLL advantage outweighs length and repetition penalties, rejecting better HF candidates. DEV favored HF reselection at 30.08% versus 30.09% and guided the fixed acquisition control.

Direct retains 67.13% of positive gain from unconditional final HF, compared with S+R's 99.98% from N+R. Relative to its own control, Direct forgoes 1.24 pp of benefit but avoids 3.58 pp of harm.

### C. Acceptance on matched HF banks

On HF banks, H4 improves over G4 but trails Direct (31.52% versus 31.46% CV27 WER). HWR4 yields the lowest complete CV27 WER of 31.44% (Table II, where S denotes each acceptor's selective policy). Triggered HR4 trails H4 in

TABLE III
POPULATION-WEIGHTED MS AND OBSERVED PEAK ALLOCATED MIB WITH BOTH MODELS RESIDENT. EACH STUDY USES THREE PAIRED ROUNDS.

| Path | FLEURS | | CV27 | |
|---|---|---|---|---|
| | ms | MiB | ms | MiB |
| *A. Four-path comparison across three checkpoints* | | | | |
| Native HF | 1291.7 | 2980.8 | 517.4 | 2813.8 |
| B | 1352.7 | 3144.9 | 591.9 | 3133.1 |
| N+R | 1436.8 | 2988.3 | 639.2 | 3133.1 |
| S+R | 1362.8 | 3189.7 | 610.0 | 3133.1 |
| *B. Seven-path comparison on one checkpoint* | | | | |
| Native HF | 1304.8 | 2978.4 | 531.2 | 2811.8 |
| Raw top-1 | 1222.6 | 2946.6 | 479.9 | 2571.0 |
| B | 1364.0 | 3142.4 | 605.1 | 3131.1 |
| N+R | 1449.6 | 2986.3 | 657.9 | 3131.1 |
| S+R | 1374.5 | 3187.2 | 628.0 | 3131.1 |
| Direct | 1372.8 | 3187.2 | 625.8 | 2811.8 |
| H3Direct | 1314.0 | 3187.2 | 586.4 | 2811.8 |
| *Relative latency saving: S+R versus N+R (95% CI)* | | | | |
| Study A | 5.15% [2.84, 9.41] | | 4.57% [4.40, 4.73] | |
| Study B | 5.18% [2.88, 9.39] | | 4.54% [4.44, 4.66] | |
| *Study B: differences between policies* | | | | |
| B − Raw top-1, ms | 141.4 | | 125.2 | |
| N+R − B, ms | 85.6 | | 52.8 | |
| S+R − B, ms | 10.5 | | 22.9 | |
| Fallback increment saved | 87.7% | | 56.6% | |

every seed ($+0.91$, $+0.08$ and $+1.17$ pp). Edit-gap weighting reverses the mean contrast to $-0.08$ pp through a 0.25 pp gain in checkpoint two that replaces 11 repetitive beams with lower-edit HF candidates. Intervals provide 95% Bonferroni coverage per family, with 10,000 synchronized draws for the later family. Using G4's language/bank-weighted RMS instead of pooled RMS for HF refits changes CV27 WER by $+0.044$, $-0.057$ and $+0.003$ pp for H4, HR4 and HWR4, respectively, without changing their ordering.

FLEURS gains round to 0.37 pp for fitted acceptors because S equals its own B off-trigger. H3's B has two fewer errors on one of ten triggers, but S errors match and the average difference is below 0.006 pp. Nontriggers account for 97.23% of the refitted H3–H4 WER gap in routine selection, without isolating PLL's causal contribution or time share.

### D. Cost, heterogeneity and supporting diagnostics

Within reranked pipelines, S+R preserves N+R's gain at lower cost (Table III). Study B's rescoring increments above raw top-1 are 141.4 ms and 125.2 ms on FLEURS and CV27. Selective acquisition saves 87.7% and 56.6% of fallback overhead. The unweighted 25-recording CV27 profile averages 85.9 ms teacher scoring and 149.3 ms PLL, including two skipped singletons. More singletons skip both stages in the population, explaining the smaller weighted increments. Native HF skips rescoring, trading faster decoding for 1.46 pp higher FLEURS WER and over 2.1 pp on CV27 TEST.

H3Direct uses H3 off-trigger and HF on triggers, omitting PLL. On CV27 it saves 6.29% over Direct but trails Native HF by 55.2 ms (95% interval 52.3 to 58.2 ms); the FLEURS interval spans zero. H4 and HWR4 only replace G4's weights; selective PLL remains future work. In Study B, Direct skips scoring on the CV27 trigger responsible for B's peak CUDA memory. On FLEURS, Direct and H3Direct match S+R's peak CUDA allocation when both windows' states are retained during recovery.

TABLE IV
TEST GAINS AVERAGED OVER SEEDS. CALLS POOL SEED–RECORDINGS.

| Language | FLEURS | | CV27 | |
|---|---|---|---|---|
| | Calls, % | Gain, pp | Calls, % | Gain, pp |
| Arabic | 0.23 | 0.64 | 0.11 | 1.40 |
| Persian | 0.27 | 1.20 | 1.15 | 9.05 |
| Polish | 0.00 | 0.00 | 0.33 | 0.30 |
| Turkish | 0.00 | 0.00 | 0.03 | 0.51 |
| Italian | 0.00 | 0.00 | 0.02 | 0.23 |
| *Empty-HF acceptance diagnostic (18 instances, 11 recordings)* | | | | |
| Acceptance minus B reversion | $S = -118$ | $D = +120$ | $I = -1995$ | |

Only HF proposes empty TEST candidates, admitted to selective banks on triggers. Under signs $(+, +, -, -)$, the centered abstention baseline outranks nonempty candidates with $a, p \le 0$, $t > 0$ and $r \ge 0$, even without repetition. The 18 acceptances in 11 CV27 recordings remove 1,995 insertions and add 120 deletions (Table IV), consistent with hallucination suppression [33] but without verified silence labels. Reverting these acceptances to B retains 91.52% of gain, or 2.10 pp at 31.88% WER, so recovery mainly reflects nonempty repairs. Refitting H4 with $e(y) = \mathbf{1}\{y = \emptyset\}$ on the same DEV folds and grid retains all 18 acceptances at 31.52% WER. One or two DEV empties per checkpoint limit calibration.

Raw top-1 to B measures routine rescoring; B to S+R isolates sparse recovery. Persian supplies 78.77% of TEST recovery gain and 84.07% on the locked panel. Excluding Persian, four-language macro gains remain 0.61 pp on CV27 TEST and 0.45 pp on the locked panel. Script encoding, acoustics and low-rank adaptation may interact, but their effects are not isolated. The 22 two-window FLEURS recordings yield 52.69% of net gain. On cached TEST banks, local S+R reduces window requests from 15 to 10 with unchanged errors. Local Direct changes WER by under 0.01 pp, although live request order may alter RNG draws.

## V. CONCLUSION

Selective acquisition matches unconditional recovery across seeds on the locked confirmation panel. It separates sparse recovery from routine rescoring to reduce recovery overhead. Retrospective controls favor expected risk over unweighted ranking on matched HF banks in every seed.

Acceptance should match fallback screening. Reselection filters unscreened draws; Direct skips trigger-time rescoring of screened HF outputs. Source-matched HWR4 yields the lowest exploratory CV27 WER, with checkpoint-dependent gains. Script-normalized gates, local routing and confidence-gated PLL could reduce acquisition and routine rescoring costs.


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